Sri Dhanavarshini K

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Professional Summary

Computer Science Engineering graduate with a strong quantitative and analytical foundation, applying descriptive, predictive, and statistical modeling techniques to derive actionable, business-relevant insights. Proficient in Python and SQL for large-scale data analysis, machine learning model development, and building retrieval-based systems using vector datastores (FAISS). Skilled in translating statistical results into clear data visualizations and presentations for both technical and non-technical audiences. Highly motivated, curious learner eager to work with enterprise-scale data systems including Databricks and PySpark.

Education

Sathyabama Institute of Science and Technology, Chennai *Expected Graduation: July 2026*
Bachelor of Engineering – Computer Science Engineering CGPA: 8.37/10

National Matriculation Higher Secondary School, Chennai *Completed: June 2022*
Higher Secondary Education Percentage: 84%

Technical Skills

Programming & Query Languages: Python, SQL, Java, C
Data Science & Machine Learning: Predictive Modeling, Statistical Analysis, Scikit-learn, TensorFlow, NLP, Pandas, NumPy
Big Data & Vector Systems: SQL/PostgreSQL, FAISS (Vector Datastore), MongoDB; familiarity with Databricks, PySpark
Visualization & Reporting: Power BI, Excel, stakeholder-facing dashboards
Tools: Git, Linux, MS Office, Postman

Work Experience

Data Analyst Intern – Skiromax Energy LLP, Chennai *February 2025 – April 2025*

- Cleaned, transformed, and prepared energy consumption datasets for analysis using Python and Pandas, ensuring data quality and readiness for modeling and reporting.
- Conducted exploratory data analysis to identify patterns, trends, and anomalies in energy usage data, supporting operational decision-making.
- Assessed data quality issues and resolved inconsistencies as part of automating recurring data processing and reporting workflows, cutting manual effort by 30%.
- Built Power BI dashboards to track KPIs and present performance trends clearly to business stakeholders.

Junior Software Engineer – Twite AI Technologies Pvt. Ltd., Chennai *April 2026 – June 2026*

- Built SQL-driven reporting dashboards and analytics views surfacing recruitment and business performance metrics for stakeholders.
- Designed relational database schemas for an Applicant Tracking System (ATS), ensuring clean, analysis-ready data structures.
- Queried and processed large datasets using FastAPI and PostgreSQL to support enterprise reporting needs.

Projects

Multi-PDF Chatbot with RAG *Python — FAISS (Vector Datastore) — RAG*

- Built a retrieval-augmented generation (RAG) system to answer natural-language queries across multiple large PDF documents, using FAISS as the vector datastore for semantic search over document embeddings.
- Combined retrieved context with an LLM to generate accurate, grounded responses.

Health Monitoring System – Predictive Modeling *Python — Scikit-learn — Tkinter*

- Built and evaluated predictive machine learning models (Logistic Regression, K-Nearest Neighbors) to classify heart disease and diabetes risk.
- Performed statistical data preprocessing, feature extraction, and model evaluation to ensure prediction quality.
- Delivered results through a user-facing interface with an integrated recommendation chatbot.

Energy Consumption Analysis & Forecasting *Python — Pandas — Scikit-learn — Power BI*

- Cleaned, transformed, and prepared raw energy consumption data, resolving quality issues and conducting EDA to identify usage patterns, seasonal trends, and anomalies.
- Performed feature engineering and hypothesis testing to identify key drivers of energy consumption.
- Built regression models to forecast consumption trends, and applied classification and clustering techniques to segment usage patterns.
- Validated performance using standard metrics (RMSE, MAE, R^2), debugging models for reproducibility, and visualized forecasts through Power BI dashboards.

NextGen AI Law Enforcement System *Python — PyTorch — GANs — Computer Vision*

- Developed an AI-assisted forensic image enhancement system, using GAN models in PyTorch to generate suspect facial images from witness descriptions.
- Integrated GFPGAN for facial restoration and Real-ESRGAN for enhancing low-quality surveillance images.